

Vikram Jaiswal

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EXPERIENCE

AI Engineer Intern

May 2025

HB Gadgets

Nagpur

- Led a team of **4 interns** to develop an **AI chatbot** for the **Parents Eye** school bus tracking application using **LangChain**
- Engineered **prompt templates** and **context-handling pipelines** to restrict responses to domain-specific queries, improving response accuracy by **15%**
- Developed backend services for **LLM orchestration**, **session memory**, and **API integration**
- Integrated chatbot with application workflows enabling queries for **bus location**, **bus proximity**, and **student boarding status**

Management Lead

Present

Robotics and Aviation Club

- Lead planning and operations for a **50+ member** technical club, coordinating robotics projects and technical activities
- Organized workshops and technical events while collaborating with faculty mentors and external partners
- Led teams to compete in **national and international robotics competitions**, securing **podium positions**

PROJECTS

Diabetes Prediction System | *Python, Scikit-learn, FastAPI, Pandas*

Live | Code

- Developed an end-to-end ML pipeline using the Pima Indians Diabetes Dataset (768 samples, 8 clinical features)
- Performed data preprocessing with median imputation and feature scaling
- Trained Logistic Regression, SVM, Decision Tree, and Random Forest models
- Trained multiple ML models and achieved **80% accuracy**, **0.86 ROC-AUC**, and **0.67 F1-score** on the diabetes prediction task.
- Selected Logistic Regression for deployment due to higher recall (**0.64**)

Automatic License Plate Recognition System | *YOLO, OpenCV, Tesseract, FastAPI*

Live | Code

- Implemented an end-to-end ALPR pipeline for vehicle plate detection and OCR using YOLO and Tesseract
- Developed a real-time image analysis pipeline on **Raspberry Pi** using OpenCV, reducing inference latency by 40% while maintaining high accuracy.
- Optimized ALPR inference pipeline, increasing throughput **10x** using ONNX quantization.
- Developed backend scripts along with **SQLite** to validate vehicle numbers and flag unauthorized entries

Diabetic Retinopathy Detection | *Python, PyTorch, CNN, Streamlit*

Live | Code

- Developed a CNN-based system for detecting diabetic retinopathy from fundus images
- Achieved **94% validation accuracy**, **0.97 recall**, and **0.94 F1-score**
- Deployed the trained model as a Streamlit web application for real-time inference

EDUCATION

Ramdeobaba College of Engineering and Management

Nagpur, Maharashtra

Bachelor of Technology **CGPA: 9.04**

June 2023 – May 2027

TECHNICAL SKILLS

Programming Languages: Python, C/C++, JavaScript, TypeScript, SQL

ML & AI: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, HuggingFace, XGBoost

Web & Backend: React.js, Next.js, Node.js, Express.js, REST APIs, JWT Authentication

Databases: MongoDB, PostgreSQL, Supabase

Robotics Tools: ROS, Gazebo

Dev Tools: Git, GitHub, Postman, VS Code, Vercel, Linux CLI